Creepage Distance

Insulator dimensioning for DC overhead lines



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Abbreviations

AC	Alternating current
CIGRE	International Council on Large Electric Systems
DC	Direct current
ESDD	Equivalent salt deposit density
HVDC	High-voltage direct current
IEC	International Electrotechnical Commission
NSDD	Non-soluble deposit density
OHL	Overhead line
SPS	Site pollution severity
TB	Technical brochure
TS	Technical specification
USCD	Unified specific creepage distance

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Creepage distance
Definition and reference voltage

Creepage distance

Shortest distance along the insulator surface between the two conductive parts that normally carry the operating voltage.

- Pollution sets it: a wetted pollution layer conducts and can flash over at normal operating voltage.
- Specified as a USCD in mm/kV of the pole-to-earth DC operating voltage [1].
- No single value: the USCD follows from site pollution and insulator type [1], [2].

$$L = \text{USCD} \cdot U_{dpe} \cdot C_a$$

L	total creepage in mm
USCD	unified specific creepage distance in mm/kV
U_{dpe}	pole-to-earth DC operating voltage in kV
C_a	altitude correction ($= 1$ up to 1000 m, see altitude slide)

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Creepage distance
Why DC needs more than AC

- The steady DC field attracts charged particles, so a DC insulator collects roughly 1.5–2 times the pollution of an AC insulator at the same site [2].
- A dry-band arc on DC has no current zero to extinguish it.
- For the same site, DC USCD values are therefore well above AC values.

Aspect	AC	DC
Pollution build-up	wind, gravity	plus electrostatic attraction
Dry-band arc	current zero every half cycle	no current zero
Ionic migration in glass and porcelain	negligible	needs DC-type glass, zinc sleeve on pin
Relative USCD	1	about 1.5–2

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Site pollution
ESDD and NSDD

- **ESDD**: mass of NaCl per insulator surface area that, dissolved in water, gives the same conductivity as the soluble deposit washed off the insulator, in mg/cm^2 [3]. The soluble deposit is the salt (sea, industrial, desert dust) that dissolves into ions when wetted and makes the layer conduct.
- **NSDD**: mass of the non-soluble residue (dust, sand, soil) per surface area, in mg/cm^2 [3]. It holds water and so helps the soluble salt to form a conducting layer.
- Both are measured by washing a reference insulator with demineralised water after a year or more on site.

$$\sigma_{20} = \sigma_\theta \cdot [1 - b \cdot (\theta - 20\,^\circ\mathrm{C})]$$

$$S_a = (5.7 \cdot \sigma_{20})^{1.03}$$

$$\mathrm{ESDD} = \frac{S_a \cdot V}{A}$$

$$\mathrm{NSDD} = \frac{1000 \cdot (W_f - W_i)}{A}$$

$\sigma_\theta, \sigma_{20}$	wash-water conductivity at θ and at 20 °C in S/m
b	temperature factor [4]
S_a	salinity in kg/m^3 (= mg/cm^3)
V, A	wash-water volume in cm^3, washed area in cm^2
W_i, W_f	filter mass before and after, dried, in g

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Site pollution
From ESDD and NSDD to creepage

- ESDD and NSDD together give the SPS class of the site, from very light to very heavy [3].
- For DC the SPS must be measured on DC-energised insulators, or estimated from AC or non-energised data with a DC/AC accumulation ratio of about 1.5–2 [1], [2].
- Flashover strength falls with ESDD and NSDD as a power law, so the USCD rises with ESDD^{a} and NSDD^{b_n}; typically $a \approx 0.2$–0.4, $b_n \approx 0.1$–0.2.

$$E_f = \frac{U_f}{L} = A_f \cdot \text{ESDD}^{-a} \cdot \text{NSDD}^{-b_n}$$

$$\text{USCD}_2 = \text{USCD}_1 \cdot \left(\frac{\text{ESDD}_2}{\text{ESDD}_1}\right)^{a} \cdot \left(\frac{\text{NSDD}_2}{\text{NSDD}_1}\right)^{b_n}$$

E_f	flashover strength per creepage in kV/m
A_f	constant of the insulator type
a, b_n	ESDD and NSDD exponents from the supplier's DC pollution tests or published data [2]

Example: doubling ESDD with $a = 0.3$ needs $2^{0.3} \approx 1.23$, i.e. 23 % more creepage.

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Creepage distance
Indicative USCD for DC OHL insulators

Site pollution	Ceramic / glass	Composite
Light	30–40 mm/kV	25–30 mm/kV
Medium	40–50 mm/kV	30–38 mm/kV
Heavy	50–60 mm/kV	38–45 mm/kV
Very heavy	mostly too long	45–55 mm/kV

- Composite (silicone) insulators keep a water-repellent surface, so they need roughly 20–30 % less creepage.
- Most recent HVDC lines therefore use composite long-rod insulators.
- The ranges are a starting point; the design value comes from the measured site pollution severity.

Indicative values from IEC TS 60815-4 [1] and CIGRE TB 518 [2]; not agreed AmWatt design constants.

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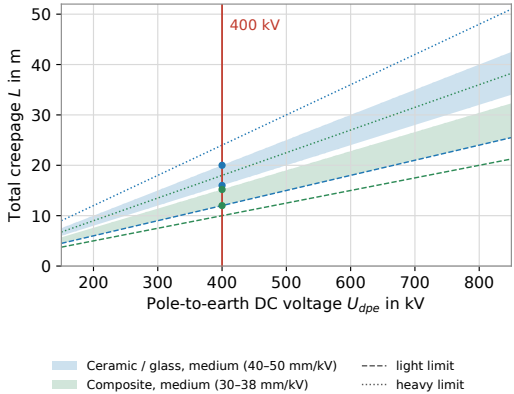
Creepage distance
Worked example at 400 kV

Study voltage $U_{dpe} = 400\,\text{kV}$, site below 1000 m ($C_a = 1$):

$$L_{cer} = 45\,\text{mm/kV} \cdot 400\,\text{kV} = 18.0\,\text{m}$$

$$L_{comp} = 34\,\text{mm/kV} \cdot 400\,\text{kV} = 13.6\,\text{m}$$

- Medium pollution, mid-range USCD for each insulator type.
- With about 600 mm creepage per DC fog-type disc, the ceramic string needs about 30 discs.
- Over the full light-to-heavy range: 12.0–24.0 m (ceramic).



Total creepage versus pole voltage, medium-pollution bands.

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Creepage distance
Altitude correction

- Lower air pressure lowers the pollution flashover voltage, so above 1000 m more creepage is needed.
- m comes from pollution tests at reduced pressure and depends on insulator type, profile and pollution level; DC tests give about 0.3–0.5 [2].
- Example: 400 kV ceramic at 2000 m, $m = 0.5$: $18.0\,\text{m} \cdot 1.063 \approx 19.1\,\text{m}$.

$$U_f = U_{f0} \cdot \left(\frac{p}{p_0}\right)^m \qquad \frac{p}{p_0} = e^{-H/8150\,\text{m}}$$

$$C_a = e^{m \cdot \frac{H - 1000\,\text{m}}{8150\,\text{m}}}$$

H	$m = 0.35$	$m = 0.5$
1500 m	1.022	1.031
2000 m	1.044	1.063
2500 m	1.067	1.096

U_f, U_{f0} pollution flashover voltage at pressure p and at p_0 (sea level); H altitude; m empirical pressure exponent.

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Creepage distance
Design procedure

From site to string length

Measure the pollution, select the USCD, correct it, then check the string against the overvoltage requirements.

1. Determine the SPS along the route: ESDD and NSDD from test insulators exposed under DC voltage, or from service experience [1], [2].
2. Convert the SPS into a USCD with the IEC TS 60815-4 / CIGRE curves, or with the supplier's DC pollution test data for the chosen insulator profile.
3. Apply the altitude correction C_a above 1000 m, and the corrections for insulator diameter and profile.
4. Check the resulting string against the switching and lightning impulse requirements; the longer of the two governs.

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References

[1] IEC TS 60815-4:2016, *Selection and dimensioning of high-voltage insulators intended for use in polluted conditions — Part 4: Insulators for d.c. systems*.

[2] CIGRE Working Group C4.303, TB 518, *Outdoor insulation in polluted conditions: Guidelines for selection and dimensioning — Part 2: The DC case*, 2012.

[3] IEC TS 60815-1:2008, *Selection and dimensioning of high-voltage insulators intended for use in polluted conditions — Part 1: Definitions, information and general principles*.

[4] IEC 60507:2013, *Artificial pollution tests on high-voltage ceramic and glass insulators to be used on a.c. systems*.

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